

Qualifying Exam in Microeconomics
Arizona State University
August 2015

There are four questions and you have four hours to complete the exam.

1. Consider the following Edgeworth box economy. Endowments are $w_1 = (2, 1)$ and $w_2 = (1, 2)$. Preferences are $u_1 = (x_{11}x_{12})^{\frac{1}{2}}$ and $u_2 = (x_{21}x_{22})^{\frac{1}{2}}$. The first subindex identifies the agent, the second one the good.

(a) Find a Walrasian equilibrium.

(b) Prove that the allocation $x_1 = (1, 1)$ and $x_2 = (2, 2)$ is Pareto optimal. The Second Welfare Theorem shows that there are transfers and prices such that the allocation is a Walrasian equilibrium with transfers. Show, without using the Second Welfare Theorem, that indeed the proposed allocation is a Walrasian equilibrium with transfers.

2. A competitive firm has a nonempty, closed production set $Y \subset \mathbb{R}^L$ for $L \geq 2$. In addition it satisfies

(a) $y \in Y$ only if $y = (q, -z)$ where $q \in \mathbb{R}_+^m$ is a vector of outputs and $z \in \mathbb{R}_+^n$ is a vector of inputs (the firm produces m outputs using n inputs); and

(b) for every $q \in \mathbb{R}_+^m$, there is a z such that $(q, -z) \in Y$ (the firm can produce any $q \in \mathbb{R}_+^m$).

Let $c(q, w)$ denote the cost function: $c(q, w) = \min_{z \geq 0} w \cdot z$ subject to $(q, -z) \in Y$, where $w \gg 0$ is the input price vector. (Under the given assumptions, a solution exists to this minimization problem.)

Prove that if Y in addition exhibits *constant returns to scale*, then the cost function is homogeneous of degree 1 in q :

$$c(\lambda q, w) = \lambda c(q, w) \quad (1)$$

for every $q \in \mathbb{R}_+^m$ and $\lambda \geq 0$. (Suggestion: consider the $\lambda > 0$ and $\lambda = 0$ cases separately.)

3. Consider a standard (but not necessarily symmetric) independent private values model (IPVM) with I agents, indexed by $i = 1 \dots I$, and one good.

Consider two direct revelation mechanisms (DRM), $\{(p_i, t_i)\}_{i=1}^I$ and $\{(p'_i, t'_i)\}_{i=1}^I$ where p_i and p'_i are probability-of-trade functions and t_i and t'_i are the transfer functions corresponding to agent i . Let (Q_i, T_i) , (Q'_i, T'_i) be the corresponding expected-probability-of-trade and expected-transfer functions for agent i in each mechanism. Let u_i and u'_i be the interim utility functions associated with the mechanisms.

- (a) Please prove that if $\forall i, Q_i = Q'_i$, then $\forall i, T_i = T'_i + [u'_i(0) - u_i(0)]$ where u_i and u'_i are the interim utility functions associated with the two mechanisms.
- (b) Briefly explain the significance of the result.

4. Consider a *finite*, two-person sequential-move game Γ of perfect information: Player 1 first chooses an action a_1 from a nonempty finite set A_1 ; after observing a_1 , player 2 chooses an action a_2 from a nonempty finite set A_2 . Player's i 's preferences over *outcomes* of this extensive form game are represented by $U_i : A_1 \times A_2 \rightarrow \mathbb{R}$ for $i = 1, 2$. Consider the following optimization problem:

$$\max_{(a_1, a_2) \in A_1 \times A_2} U_1(a_1, a_2) \quad (2)$$

subject to $a_2 \in \operatorname{argmax}_{a'_2 \in A_2} U_2(a_1, a'_2)$.

Evaluate the assertions in parts (a) and (b). If the assertion is true, then prove it. If it is false, then give a counterexample, identify additional conditions that make the assertion true, and prove your amended assertion. Additional conditions could be either on the primitives (the U_i 's and A_i 's) or directly on the best replies.

- (a) $(a_1^*, a_2^*) \in A_1 \times A_2$ is a subgame perfect Nash equilibrium *outcome* of Γ if it solves problem (2).
- (b) $(a_1^*, a_2^*) \in A_1 \times A_2$ is a subgame perfect Nash equilibrium outcome of Γ *only* if it solves problem (2).
- (c) Now drop the assumption that Γ is finite. How is the truth of the assertions of (a) and (b) affected? Explain briefly.

$\Rightarrow$ SPNE $\Rightarrow$ (2) (b) no restriction

$\Rightarrow$ (2) $\Rightarrow$ SPNE

Solution to (2) might not exist!

Principal-agent assumes agent does what is good for principal when indifferent - but that is not always true.

$\Rightarrow$ infinite # of actions